

CSCI E-278 - Assignment 4 Problem Set Solutions

Question 1

An analyst begins with a large set of fundamental factors, many of which are correlated. After applying principal component analysis (PCA), the analyst obtains a much smaller set of orthogonal principal components that retain most of the variation in the original data. Explain what PCA has accomplished. If one principal component explains more of the variance in the feature data than another, why does that not necessarily mean it will be more useful for predicting a particular investment outcome? How can the component loadings help the analyst interpret what a principal component represents?

Answer. PCA replaces correlated original features with a smaller set of mutually orthogonal linear combinations while preserving as much variation in the feature data as possible. A component that explains more feature variance is not automatically more predictive because PCA is unsupervised: it maximizes variance in X without considering the investment target y . A lower-variance component can therefore contain more information about the target. The loadings show how strongly each original variable contributes to a component; large positive or negative loadings help identify the economic theme or factor the component represents.

Question 2

Why should an analyst define the prediction target and forecast horizon before engineering features or choosing a model?

Answer. The target and horizon determine what information is actually available at prediction time and what relationship the model is supposed to learn. A feature useful for next-day returns may be irrelevant for a six-month outcome, while poorly aligned timestamps can introduce look-ahead bias. Defining y and the horizon first keeps feature construction, lags, evaluation metrics, and validation aligned with the real trading decision.

Question 3

What should an analyst look for in a residual plot, and what residual patterns would suggest that a model is misspecified?

Answer. A healthy residual plot should look roughly random around zero with no systematic structure and reasonably stable spread. Trends, curvature, clusters, changing variance, or serial correlation suggest the model has left predictable structure unexplained. Those patterns can indicate missing nonlinearities, omitted variables, heteroscedasticity, time dependence, or regime effects.

Question 4

Why is ordinary shuffled K-fold cross-validation often inappropriate for financial time-series prediction? Describe a better validation design and explain how regime changes affect evaluation.

Answer. Randomly shuffling observations can place future data in the training set and earlier data in validation, creating look-ahead leakage and overstating performance. Time-series evaluation should preserve chronology, using expanding-window or rolling-window splits in which training always precedes validation. Because market regimes change, performance should also be checked across multiple time periods; a model that works only in one regime may not generalize.

Question 5

Suppose Model A has a lower AIC than Model B on the training data, but Model B performs materially better on a truly unseen chronological test set. Which model should an analyst prefer for prediction, and why?

Answer. For a forecasting objective, the analyst should generally prefer Model B because genuine out-of-sample performance is the more direct test of predictive generalization. AIC balances in-sample likelihood against model complexity and is useful for model comparison under its assumptions, but it does not override evidence from a properly designed unseen test set. The key is that the test set must remain untouched during model selection.